Q. Can you tell us about some of the latest technologies that have the potential to transform the open source industry in the near future.

A.The enterprises' journey of migration from monolith to microservices is an exciting trend in the open source environment. Containers and Kubernetes play key roles in enabling this migration, with Linux underpinning the latter. Automation is another interesting trend gaining traction in the industry. Fuelled by enterprise DevOps adoption, the automation of software delivery and infrastructure changes gives developers the freedom to concentrate on coding without worrying about infrastructure agility, scalability, etc. The other trend I see gaining momentum is the concept of the service mesh. This is essentially platform-level automation for creating the network connectivity required by microservices-based software architectures.

Q. How has open source evolved to become the new norm for Indian enterprises, especially with technologies like AI, ML and IoT increasingly coming into play?

A. Enterprise IT environments require a lot of investment and planning, as they need IT infrastructure that delivers speed, scalability, flexibility and faster innovation to help the business flourish. Open source provides all this, along with a much-needed predictable life cycle. It combines the best of two worlds — the advantages of having access to the source code, along with the stability, performance and support that is offered by enterprise software, essentially making open source the new norm for enterprises.

Today, the industry is witnessing projects that are trying to make AI and ML more accessible to software developers. Take the case of Kubeflow, an ML toolkit for Kubernetes. The idea behind Kubeflow is to make it simple to scale machine learning models and deploy them to production wherever Kubernetes is running. Once again, we see Kubernetes becoming the target platform of choice.

Q. What are the new trends in the Indian developer community today?

A. Developers around the world continue to have a keen inclination towards open source, as open code gives them a deeper understanding of the technology and helps them in contributing to the community, while increasing cost-effectiveness and efficiency at the same time. They also want to keep abreast with the latest technologies such as AI, ML, blockchain, etc. Python, JavaScript, PHP, C and C++ are still some of the programming languages with the highest demand and uptake. Developers are also foraying into containerisation, the entire Dockers and Kubernetes ecosystem and serverless microservices.

> "At Red Hat, we strongly believe in the power of participation and innovation through collaboration."

Q. What are the challenges developers face when deploying applications on the cloud, and how is Red Hat helping them overcome these?

A. To meet evolving business demands, IT organisations are looking at new workloads, AI, ML and IoT to drive competitive advantages in crowded marketplaces. As a result, the hybrid cloud and multi-cloud are becoming the preferred architecture for enterprises. However, developers face several challenges while making an app cloud-native. The most important challenge is interoperability and the support issues when migrating services from one cloud provider to another.

For more than 15 years, Red Hat has helped enterprises innovate on Linux, first in their data centres and now across the hybrid cloud via Red Hat Enterprise Linux (RHEL 8). For a workload running on any environment, RHEL 8 delivers a Linux experience to meet the unique technology needs of evolving enterprises. It delivers the innovative muscle along with a hardened code base, extensive security updates, award-winning support, and a vast ecosystem of tested and validated supporting technologies. It offers several of the following benefits to developers:

- It simplifies application development – with less setup and configuration effort, you can more quickly get to writing code.
- It is for traditional and cloud/container applications, with many new tools for both.
- It offers dozens of tools to build and test applications.

Q. What are the skillsets that are required for a developer to get hired in an open source based company like Red Hat?

A. The pillars of open source are community, transparency and collaboration. Therefore, to work in an open environment, it is important to possess in-depth technical knowledge, good communication skills, a problem-solving attitude and a passion for innovation. In open source development, in particular, developers need to articulate what they want to build, how they are building it, and how and why they have changed the existing code. This means writing clear, detailed documentation for their programs, as well as being able to talk through